

Coenzyme A Biosynthesis from Pantothenate in *Pseudomonas putida* KT2440 — Module/Pathway/Taxon Review

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00770 "Pantothenate and CoA biosynthesis" **Module area:** cofactors_vitamins_redox **Scope of this review:** the five-step conversion of pantothenate (vitamin B5) → 4'-phosphopantothenate → 4'-phosphopantothenoyl-cysteine → 4'-phosphopantetheine → dephospho-CoA → coenzyme A.

1. Executive Summary

Coenzyme A biosynthesis from pantothenate is **fully satisfiable (module status: COVERED)** in *Pseudomonas putida* KT2440. Every one of the five canonical steps is encoded by a structurally verified, correctly sized, single-copy gene in proteome UP000000556. The critical curation point is that the pathway must be described in the **bacterial enzyme frame (CoaX/CoaBC/CoaD/CoaE)**, not the human PANK1/2/3 + PPCS + PPCDC + COASY frame supplied in the generic module context. The mammalian gene names do not have one-to-one orthologs in this organism; forcing the human frame onto KT2440 would incorrectly split step 2+3 (fused into one bacterial CoaBC protein) and incorrectly fuse steps 4+5 (which are two separate bacterial proteins, not a single COASY).

The mapping is unambiguous: pantothenate phosphorylation is carried out by a **CoA-insensitive Type III pantothenate kinase, coaX/PP_0438** (EC 2.7.1.33) — not the *E. coli* Type I CoaA. The cysteine-conjugation and decarboxylation steps (the roles of human PPCS and PPCDC) are performed by a **single bifunctional CoaBC protein, dfp/PP_5285** (EC 6.3.2.5 + EC 4.1.1.36), whose two Pfam domains (Flavoprotein + DFP) each contribute one catalytic activity. The final two steps that mammalian COASY fuses are here **two separate enzymes: coaD/PP_5123** (phosphopantetheine adenylyltransferase, PPAT, EC 2.7.7.3) and **coaE/PP_0631** (dephospho-CoA kinase, DPCK, EC 2.7.1.24).

Two actionable curation flags emerged. First, **aspartate 1-decarboxylase (panD, EC 4.1.1.11) is genuinely absent** in KT2440 (confirmed by three independent negative queries including Pfam PF02261), even though it is present in *P. aeruginosa* and *P. fluorescens*. This is a strain-specific gap in *de novo* β -alanine supply; β -alanine (needed upstream for pantothenate synthesis) most likely comes from reductive pyrimidine catabolism (the *pyd* genes plus the amidohydrolases hyuC/PP_4034 and PP_0614). Second, **EC 1.1.1.169 (ketopantoate reductase, panE) is over-propagated**: domain architecture shows only **PP_1351 is the true panE**, while PP_2325 and PP_2998 are KPR-related paralogs in a distinct InterPro family and **PP_4452 (not in the supplied candidate list) is a mis-annotated opine dehydrogenase**. Finally, roughly half of the 24 supplied candidate genes belong to neighboring KEGG maps (branched-chain amino acid biosynthesis ppu00290/ppu00660, β -alanine/pyrimidine catabolism ppu00410, CoA utilization rather than synthesis) and should be excluded from the ppu00770 CoA-biosynthesis core.

2. Target-Organism Pathway Definition

What is included

This module covers the **universal five-step conversion of pantothenate to coenzyme A**:

1. Pantothenate + ATP $\rightarrow$ 4'-phosphopantothenate (pantothenate kinase; rate-limiting/feedback step in many organisms)
2. 4'-phosphopantothenate + L-cysteine + CTP/ATP $\rightarrow$ 4'-phosphopantotheno-L-cysteine (phosphopantotheno-L-cysteine synthetase; CoaB/PPCS)
3. 4'-phosphopantotheno-L-cysteine $\rightarrow$ 4'-phosphopantetheine + CO₂ (phosphopantotheno-L-cysteine decarboxylase, FMN-dependent; CoaC/PPCDC)
4. 4'-phosphopantetheine + ATP $\rightarrow$ dephospho-CoA (phosphopantetheine adenyltransferase, PPAT; CoaD)
5. Dephospho-CoA + ATP $\rightarrow$ CoA (dephospho-CoA kinase, DPCK; CoaE)

What must be kept separate (neighboring pathways / overview maps)

- **Upstream pantothenate (vitamin B5) biosynthesis** — *panB*, *panC*, *panD*, *panE*. KEGG lumps these into the same map ppu00770, but for the CoA-biosynthesis *module* proper

(pantothenate → CoA) they are the supply route, not the core. They should be tracked as an adjacent sub-module.

- **Branched-chain amino acid biosynthesis (ppu00290, ppu00660; valine/leucine/ isoleucine)** — *ilvC, ilvD, ilvE, ilvH, ilvI, PP_1157, PP_1394*. These produce the 2-oxoisovalerate precursor feeding *panB*, but are not CoA-biosynthesis genes.
- **β-alanine / pyrimidine catabolism (ppu00410)** — *pydA/B/X (PP_4036–4038), hyuC/PP_4034, PP_0614*. These supply β-alanine; relevant to pantothenate supply, not to the CoA core.
- **CoA utilization (not synthesis)** — the 4'-phosphopantetheinyl transferase **entD/PP_1183** (siderophore/PKS holo-ACP loading), **acpH/PP_0922** (ACP phosphodiesterase, removes phosphopantetheine), and **mazG/PP_1657** (NTP pyrophosphohydrolase / house-cleaning). These consume or recycle CoA-derived groups and are not part of the biosynthetic path.

Alternate names / database definitions

- **KEGG** ppu00770 "Pantothenate and CoA biosynthesis" merges vitamin B5 synthesis and CoA synthesis into one map.
 - **Bacterial gene nomenclature:** *coaA* (Type I PanK), *coaX* (Type III PanK), *coaBC/dfp* (bifunctional PPCS-PPCDC), *coaD* (PPAT), *coaE* (DPCK).
 - **Human/eukaryotic frame:** PANK1/2/3 (three PanK isoforms), PPCS, PPCDC, COASY (bifunctional PPAT+DPCK). The generic module context supplied uses this frame; it does **not** map cleanly onto *Pseudomonas*.
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3. Expected Step Model (in the bacterial frame)

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                β-alanine (from pyrimidine catabolism: pyd + hyuC/PP_0614;
                        panD route ABSENT in KT2440)
                |
2-oxoisovalerate | (panB PP_4699, KPHMT)
                |   v
ketopantoate --(panE PP_1351, EC 1.1.1.169)--> pantoate --(panC PP_4700, EC 6.3.2.1)--> PAN

===== CoA-biosynthesis CORE (this module) =====

STEP 1  pantothenate  --[coaX / PP_0438, Type III PanK, EC 2.7.1.33]-->  4'-phosphopantothenate
STEP 2  + L-cysteine  --[dfp / PP_5285 CoaB domain (DFP, PF04127), EC 6.3.2.5]--> P-pantothenate
STEP 3                                     --[dfp / PP_5285 CoaC domain (Flavoprotein, PF02441), EC 4.1.1.36]--> 4-phosphopantothenate
STEP 4                                     --[coaD / PP_5123, PPAT, EC 2.7.7.3]-->  dephospho-CoA
STEP 5                                     --[coaE / PP_0631, DPCK, EC 2.7.1.24]-->  COENZYME A

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Key structural notes vs. the human frame: - Human PANK1/2/3 (three feedback-inhibited isoforms) → **one** *Pseudomonas* Type III PanK (CoaX), **not** feedback-inhibited by CoA/acyl-CoA. - Human PPCS + PPCDC (two proteins) → **one** bifunctional CoaBC (dfp) protein. - Human COASY (one bifunctional PPAT+DPCK protein) → **two** separate *Pseudomonas* proteins (CoaD + CoaE).

4. Candidate Genes and Evidence

4.1 Core CoA-biosynthesis genes (COVERED)

Step	EC	KT2440 gene	Locus / UniProt	Length	Domain evidence	Confidence
1. Pantothenate kinase	2.7.1.33	coaX	PP_0438 / Q88QQ0	249 aa	PF03309 Pan_kinase; InterPro Type_III_PanK + ATPase_NBD (ASKHA fold)	High
2. PPCS (CoaB)	6.3.2.5	dfp (CoaBC)	PP_5285 / Q88C96	403 aa	PF04127 DFP domain	High
3. PPCDC (CoaC)	4.1.1.36	dfp (CoaBC)	PP_5285 / Q88C96	403 aa	PF02441 Flavoprotein domain	High
4. PPAT	2.7.7.3	coaD	PP_5123 / Q88CQ7	161 aa	PF01467 CTP_transf_like; InterPro PPAT	High
5. DPCK	2.7.1.24	coaE	PP_0631 / Q88Q65	207 aa	PF01121 CoaE; InterPro Depp_CoAkinase + P-loop_NTPase	High

coaX / PP_0438 (Type III pantothenate kinase). Proteome UP000000556 contains a single EC 2.7.1.33 pantothenate kinase, annotated "Type III pantothenate kinase (PanK-III)." No Type I (*coaA*) or Type II PanK is present. This is biologically expected: Type III PanK is documented as the **only** PanK in the close relative *Pseudomonas aeruginosa*. A curation-relevant caveat is that Type III PanK is **not inhibited by CoA or its thioesters**, so the generic module's "rate-limiting and feedback-regulated" language (true for human PANK1/2/3) **does not transfer** to KT2440. Domain architecture (PF03309 Pan_kinase, ASKHA/ATPase fold) confirms a structurally intact

enzyme at 249 aa. *Evidence: direct UniProt annotation for KT2440 + strong transfer from P. aeruginosa*

(P 16855243).

dfp / PP_5285 (bifunctional CoaBC). This single 403-aa protein carries **both** catalytic domains — PF04127 (DFP; the CoaB/PPCS ligase, EC 6.3.2.5) and PF02441 (Flavoprotein; the CoaC/PPCDC decarboxylase, EC 4.1.1.36) — matching full-length *E. coli* Dfp (~406 aa). InterPro terms "CoaBC," "CoaB-like_sf," and "DNA/pantothenate-metab_flavo_C" are all present. This confirms that KT2440 fuses module steps 2 and 3 into one gene, exactly as is standard for the vast majority of prokaryotes. Curation implication: **steps 2 and 3 are both satisfied by one locus**; do not create a "gap" for a missing standalone PPCDC. *Evidence: direct UniProt/InterPro for KT2440; mechanism supported by*

P 33420031.

coaD / PP_5123 (PPAT). Single EC 2.7.7.3 enzyme, 161 aa, PF01467 (CTP_transf_like), InterPro "PPAT." Correctly sized nucleotidyltransferase. High confidence. *Direct KT2440 annotation.*

coaE / PP_0631 (DPCK). Single EC 2.7.1.24 enzyme, 207 aa, PF01121 (CoaE), InterPro "Depp_CoAkinase" + "P-loop_NTPase." High confidence. *Direct KT2440 annotation.*

4.2 Upstream pantothenate-supply genes (adjacent sub-module, not CoA core)

Enzyme	EC	Gene	Locus / UniProt	Status
Ketopantoate hydroxymethyltransferase (KPHMT)	2.1.2.11	panB	PP_4699 / Q88DW9	Present, high confidence
Ketopantoate reductase	1.1.1.169	panE	PP_1351 / Q88N64	True panE (see §5)
Pantothenate synthetase	6.3.2.1	panC	PP_4700 / Q88DW8	Present, high confidence
Aspartate 1-decarboxylase	4.1.1.11	panD	—	ABSENT (see §5)

panE / PP_1351. InterPro cross-references for this 305-aa protein are **identical** to experimentally characterized *E. coli* PanE (P0A9J4): IPR050838 (Ketopantoate_reductase), IPR003710 (ApbA), IPR013752 (KPA_reductase), IPR013332 (KPR_N). This is the true ketopantoate reductase. High confidence.

4.3 Genes belonging to neighboring pathways (exclude from ppu00770 CoA core)

Gene(s)	Locus	Native pathway	Reason to exclude
ilvC, ilvD, ilvE, ilvH, ilvI	PP_4678, PP_5128, PP_3511, PP_4679, PP_4680	BCAA biosynthesis (ppu00290/00660)	Provide 2-oxoisovalerate precursor; not CoA genes
PP_1157, PP_1394	—	Acetolactate synthase (ppu00660)	BCAA branch
pydA/B/X	PP_4038/4036/4037	Pyrimidine catabolism (ppu00410)	β-alanine supply, catabolic
hyuC, PP_0614	PP_4034, PP_0614	β-alanine amidohydrolases (ppu00410)	β-alanine supply
entD	PP_1183	Siderophore/PKS (ppu00074)	4'-phosphopantetheinyl transferase — CoA <i>utilization</i>
acpH (PP_0922)	PP_0922	ACP recycling	Removes phosphopantetheine — CoA <i>utilization</i>
mazG	PP_1657	Nucleotide house- cleaning	NTP pyrophosphohydrolase — not CoA synthesis

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 GAP — panD (aspartate 1-decarboxylase, EC 4.1.1.11) is genuinely absent

Three independent negative queries against proteome UP000000556 returned zero hits: (i) no protein carrying the aspartate-1-decarboxylase Pfam domain **PF02261**, (ii) no gene named *panD*, (iii) no protein with EC 4.1.1.11. A name search for "aspartate decarboxylase" returned only unrelated decarboxylases (*gcvP1/2*, *ldcC*, *cobC*). By contrast, *panD*/EC 4.1.1.11 is annotated in *P. aeruginosa* PAO1 (Q9HV68), PA14 (Q02FU1), LESB58 (B7V1E3), PA7 (A6VCI7), and *P. fluorescens* (Q848I5).

This is a **strain-specific gap in the canonical *de novo* β -alanine route** (aspartate $\rightarrow$ β -alanine + CO₂). Because β -alanine is required to make pantothenate (*panC* condenses pantoate + β -alanine), KT2440 must obtain β -alanine another way. The most parsimonious alternative is **reductive pyrimidine catabolism**, which is fully encoded in KT2440: *pydB* dihydropyrimidinase (EC 3.5.2.2, PP_4036), *pydA/pydX*, and the N-carbamoyl- β -alanine amidohydrolases *hyuC*/PP_4034 and PP_0614 (EC 3.5.1.6). Curation action: mark the *panD* step **not covered by a *panD* ortholog** but flag β -alanine supply as **satisfied by an alternative route (pyrimidine catabolism)** — i.e., a `candidate_uncertain`/lineage-specific-alternative for β -alanine supply rather than a hard pathway break.

5.2 OVER-ANNOTATION — EC 1.1.1.169 propagated to four loci; only one is *panE*

Four KT2440 loci carry EC 1.1.1.169 (2-dehydropantoate/ketopantoate reductase): PP_1351, PP_2325, PP_2998, and PP_4452 (the last not in the supplied candidate list). InterPro domain benchmarking against *E. coli* PanE (P0A9J4) resolves the ambiguity decisively:

Locus	UniProt	Length	InterPro panel	Verdict
PP_1351	Q88N64	305 aa	IPR050838 + IPR003710 + IPR013752 + IPR013332 (identical to <i>E. coli</i> PanE)	True panE
PP_2325	Q88KG5	315 aa	IPR003710 (ApbA) but IPR051402 (KPR-Related); lacks IPR050838	KPR-related paralog
PP_2998	Q88IK1	315 aa	IPR003710 (ApbA) but IPR051402 (KPR-Related); lacks IPR050838	KPR-related paralog
PP_4452	Q88EL0	359 aa	No ApbA/KPR signature; IPR003421 (Opine_DH) + IPR051729 (Opine/Lysopine_DH)	Mis-annotated opine dehydrogenase

Curation action: retain EC 1.1.1.169 / panE only on **PP_1351**. Demote PP_2325 and PP_2998 to "ketopantoate-reductase-related (KPR family, uncharacterized specificity)." **Correct PP_4452's annotation** — it is an opine/lysopine dehydrogenase, not a ketopantoate reductase; its EC 1.1.1.169 assignment is a clear over-propagation error.

5.3 Frame mismatch — human PANK/PPCS/PPCDC/COASY names do not map 1:1

The generic module supplies the human gene frame. In KT2440: (a) there is one PanK, not three, and it is CoA-insensitive Type III; (b) PPCS+PPCDC are one gene (dfp), not two; (c) COASY is split into two genes (coaD + coaE). Any module document that requires named PANK1/2/3, PPCS, PPCDC, and COASY orthologs would spuriously report gaps. The module must be defined at the **activity/EC level**, not the human-gene level.

6. Module and GO-Curation Recommendations

Module step dispositions

Module step	Activity (EC)	KT2440 locus	Disposition
Step 1 — pantothenate phosphorylation	2.7.1.33	coaX / PP_0438	covered (Type III PanK; note: not feedback-inhibited)
Step 2 — cysteine conjugation	6.3.2.5	dfp / PP_5285 (DFP domain)	covered
Step 3 — decarboxylation	4.1.1.36	dfp / PP_5285 (Flavoprotein domain)	covered
Step 4 — adenylation	2.7.7.3	coaD / PP_5123	covered
Step 5 — phosphorylation to CoA	2.7.1.24	coaE / PP_0631	covered
(Adjacent) β -alanine supply — panD	4.1.1.11	none	gap / lineage-specific alternative (pyrimidine catabolism)

Overall module status: COVERED (all five CoA-core steps satisfied).

Boundary / definition recommendations

- **module_needs_revision (framing):** replace the human PANK1/2/3 + PPCS + PPCDC + COASY frame with the bacterial CoaX/CoaBC(dfP)/CoaD/CoaE frame for this taxon. Define steps by EC/activity so the one-PanK, fused-CoaBC, split-CoaD/CoaE bacterial architecture validates as complete.
- **Keep the CoA-biosynthesis core separate** from the vitamin B5 supply sub-module (panB/C/D/E) and from BCAA biosynthesis and pyrimidine/ β -alanine catabolism, even though KEGG ppu00770 merges the first two.
- **CoA-utilization genes** (entD/PP_1183, acpH/PP_0922, mazG/PP_1657) should be removed from the ppu00770 core candidate set.

GO-curation notes

- **coaX/PP_0438**: GO:0004594 (pantothenate kinase activity) — appropriate. Do **not** attach CoA-feedback-inhibition regulatory annotations transferred from human PANK.
- **dfp/PP_5285**: annotate **both** GO:0004632 (phosphopantothenate–cysteine ligase) and GO:0004633 (phosphopantothenoylcysteine decarboxylase) on the one protein.
- **PP_2325/PP_2998**: broaden/soften from "ketopantoate reductase (panE)" to KPR-family; avoid propagating EC 1.1.1.169 as a confirmed function.
- **PP_4452**: request re-annotation to opine/lysopine dehydrogenase (IPR003421); remove EC 1.1.1.169.
- No new GO term requests appear necessary; existing terms cover all activities.

7. Genes to Promote to Full fetch-gene Review

1. **dfp / PP_5285 (CoaBC)** — highest priority. Confirm both catalytic domains are intact and that the single locus is the only route for steps 2+3 (no standalone PPCDC backup). Central, single-copy, essential.
 2. **coaX / PP_0438** — confirm Type III assignment and CoA-insensitivity phenotype; document that no Type I/II PanK exists as backup.
 3. **PP_4452** — promote to correct the opine-dehydrogenase mis-annotation and strip the erroneous EC 1.1.1.169/panE label.
 4. **PP_2325 and PP_2998** — review to soften panE/EC 1.1.1.169 over-propagation and assign accurate KPR-family descriptions.
 5. **β-alanine supply cluster (hyuC/PP_4034, PP_0614, pydA/B/X PP_4036–4038)** — review as the likely compensatory route for the absent panD, to formally close the β-alanine gap.
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8. Evidence Base

Finding	Evidence type	Strength / transfer
All 5 CoA steps encoded by single-copy genes	Direct UniProt annotation of KT2440 proteome UP0000000556	Direct, strong
coaX is the sole PanK and is Type III	Direct KT2440 annotation + <i>P. aeruginosa</i> literature	Direct + strong Pseudomonas transfer
dfp is a genuine two-domain CoaBC	Direct UniProt Pfam/InterPro (PF04127 + PF02441) for PP_5285	Direct, strong
coaD/coaE structurally intact	Direct UniProt Pfam/InterPro	Direct, strong
panD absent	Three independent negative proteome queries (PF02261, gene, EC)	Direct negative, strong
panD present in other Pseudomonas	UniProt annotations for PAO1/PA14/LESB58/PA7/ <i>P. fluorescens</i>	Direct for those strains
EC 1.1.1.169 over-propagation	InterPro domain benchmarking vs <i>E. coli</i> PanE (P0A9J4)	Direct comparative, strong

Literature:

- **PMID: 16855243** — *Crystal structure of a type III pantothenate kinase: insight into the mechanism of an essential coenzyme A biosynthetic enzyme universally distributed in bacteria.* Establishes that "This type III PanK, or PanK-III, is widely distributed in the bacterial kingdom and accounts for the only known PanK in many pathogenic species, such as *Helicobacter pylori*, *Bordetella pertussis*, and *Pseudomonas aeruginosa*," and that it "is not inhibited by CoA or its thioesters." Supports the assignment of PP_0438 as the sole, CoA-insensitive Type III PanK in *P. putida* and directly refutes transfer of the human feedback-regulation model to this taxon.
- **PMID: 33420031** — *Inhibiting Mycobacterium tuberculosis CoaBC by targeting an allosteric site.* States that "The biosynthesis of CoA is performed in five steps, with the second and third steps being catalysed in the vast majority of prokaryotes, including *M. tuberculosis*, by a single bifunctional protein, CoaBC." Directly supports the interpretation that KT2440 steps 2 and 3 are both covered by the single bifunctional dfp/PP_5285 protein, and that no separate PPCDC gene is expected.

9. Limitations and Knowledge Gaps

- **No direct KT2440 biochemistry:** all step assignments rest on genome/proteome annotation and domain architecture, not on purified-enzyme assays from strain KT2440 itself. Enzymatic activity is inferred, though the domain evidence is strong.
- **coaX kinetics and feedback status** are transferred from *P. aeruginosa* Type III PanK; not experimentally verified for the KT2440 protein specifically. Transfer is strong (close relative, same enzyme family) but remains inference.
- **β -alanine supply route is inferred:** the pyrimidine-catabolism hypothesis for compensating the absent panD is based on gene presence, not on demonstrated flux. It is plausible that KT2440 acquires β -alanine from the environment or from another undetermined route.
- **PP_2325 / PP_2998 substrate specificity** is unresolved — they are KPR-family but their true physiological substrates are uncharacterized.
- **Essentiality** of each locus in KT2440 was not tested here; CoA biosynthesis is essential in bacteria generally, but strain-specific transposon-fitness data were not incorporated.

10. Proposed Follow-up Experiments / Actions

1. **Close the β -alanine gap:** query KT2440 transposon-insertion fitness / RB-TnSeq datasets for *pydB* (PP_4036), *hyuC* (PP_4034), and PP_0614 under minimal medium without β -alanine/pantothenate — auxotrophy or fitness defect would confirm the compensatory route for the missing panD.
2. **Correct annotations:** submit re-annotation for PP_4452 (opine dehydrogenase, remove EC 1.1.1.169) and soften PP_2325/PP_2998 to KPR-family.
3. **Confirm dfp bifunctionality:** heterologous expression + coupled assay of PP_5285 to verify both CoaB and CoaC activities reside in the single protein.
4. **Verify coaX CoA-insensitivity** biochemically for the KT2440 enzyme to formally retire the human feedback-regulation annotation for this taxon.
5. **Module document update:** rewrite the ppu00770 CoA-core module in EC/activity terms with the bacterial CoaX/CoaBC/CoaD/CoaE frame, splitting off the vitamin-B5 supply sub-

module and excluding CoA-utilization genes (entD, acpH, mazG) and BCAA/pyrimidine-catabolism genes.

Prepared for manual module satisfiability and gene-annotation curation. Evidence is explicitly labeled as direct (KT2440 proteome/InterPro) vs. transferred (Pseudomonas/generic bacterial). Overall determination: CoA biosynthesis from pantothenate is COVERED in P. putida KT2440, with a panD/ β -alanine-supply lineage-specific alternative and an EC 1.1.1.169 over-annotation as the two actionable curation flags.

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